

Software Engineering Lab 03

Component Modelling & Architectural Pattern Selection

Problem Statement #06: Alumni Mentorship & Mock Interview Platform

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1. Architectural Choice

We chose Layered Architecture for the Alumni Mentorship & Mock Interview Platform.

The system is organized into Presentation, Business and Data layers, separating user interaction, application functionality and persistent data.

2. Reasons for Choosing Layered Architecture

Reason 1 – Separation of Responsibilities:

The platform has distinct functionality for mentor discovery and matching, availability management, session booking, interview feedback and scorecards. These responsibilities can be separated into focused business components while keeping the user interface and data storage independent.

Reason 2 – Maintainability:

The requirements from Lab 1 and the Agile backlog from Lab 2 naturally divide the system into functional areas. Changes to mentor matching, availability, booking, calendar and meeting support, or scorecard management can therefore be handled within their respective components without mixing all responsibilities together.

3. Security Advantage

Layered Architecture separates the user interface from direct access to stored data. Business components provide controlled access to user profiles, mentor availability, booked sessions and interview scorecards. This supports the privacy requirement from Lab 2, including soft deletion of user profile and scorecard data within 24 hours of a request.

4. Performance Benefit

The layered organization separates application processing from data access and allows frequently used operations such as mentor search, mentor recommendations and availability checking to be handled through dedicated components. This supports the Lab 2 requirement for search results to be returned within 2 seconds.

5. Component Organization

The Presentation Layer contains the User Interface. The Business Layer contains Authentication Service, Mentor Matching Service, Booking Manager, Calendar & Notification Service and Scorecard Manager. The Data Layer contains the Platform Database for persistent platform records.